

Sensors in Robotic Systems

Sensors serve as the primary means by which robots perceive and gather information about their surroundings, playing a critical role in enabling autonomous decision-making and interaction with the physical world. Proprioceptive sensors measure internal states of the robot itself, such as joint angles, velocities, and motor currents, providing essential feedback for control systems to accurately execute intended movements. Encoders, commonly attached to motor shafts, measure rotational position and speed, allowing precise control over joint movements in robotic arms and mobile robot wheels. Exteroceptive sensors, in contrast, gather information about the external environment, including cameras, which capture visual information used for tasks such as object recognition, navigation, and quality inspection in manufacturing settings. LIDAR sensors, which measure distances by emitting laser pulses and measuring the time for reflected light to return, are widely used in autonomous vehicles and mobile robots for mapping environments and detecting obstacles with high precision. Ultrasonic sensors, which measure distance using sound waves, offer a simpler and more cost-effective alternative for basic obstacle detection, though generally with lower precision compared to LIDAR. Force and torque sensors allow robots to measure the physical forces they apply or experience during interactions with objects or surfaces, enabling delicate manipulation tasks such as assembly or handling fragile items without causing damage. Inertial measurement units, combining accelerometers and gyroscopes, provide information about a robot's orientation and movement, essential for maintaining balance in legged robots or stabilizing drones during flight. Sensor fusion, the process of combining data from multiple sensors, is often employed to improve the accuracy and reliability of a robot's perception of its environment, compensating for the limitations and noise inherent in any single sensor type.